Frank Brase
Senior Software Architect – Hands-on Embedded, GenAI & UI Systems
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Summary
Hands-on Software Architect with 15+ years of experience in embedded and distributed systems, focusing on C++, UI architectures (Qt/QML, Flutter, GTK4) and Yocto-based Linux platforms. Combines architectural responsibility with active development, scalable SDX-based systems and GenAI-assisted rapid prototyping.

Professional Experience

Software Architect (Hands-on)
ITK Engineering, Holzkirchen | 12/2022 – 07/2026

- Design and hands-on implementation of software architectures and UI-intensive embedded systems for high-end consumer and industrial products
- Definition and implementation of system concepts: SDX, microservices, cloud-native architectures, containers, Zephyr RTOS
- GenAI-assisted rapid prototyping using GitHub Copilot, Cursor and Claude Code
- Technical enablement through architecture training, internal talks and presales support in connectivity and telemetry

Embedded Software Developer
ITK Engineering, Munich | 02/2014 – 12/2022

- Hands-on development in C++, C, Python, Rust, Go, Java and C# across customer projects
- Project domains: automotive, medical robotics and aerospace
- Visualization and telemetry of vehicle and device data on embedded Linux platforms
- UI development using Qt/QML, GTK4, Flutter and React
- Mentoring and technical coaching for internal C++ programs

Software Developer
Alcatel-Lucent (now Nokia), Nuremberg | 07/2000 – 01/2014

- Software development for mobile base stations in high-performance environments
- Integration of emerging network stacks for first-call solutions in international expert teams

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Core Skills
Software Architecture (iSAQB-F) · Embedded Linux · C++23 · Python · Rust · Go · Java · C#
UI & HMI Architectures (Qt/QML, Flutter, GTK4) · Distributed & Cloud-Connected Systems · GenAI-assisted Development

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Education
Diploma in Mathematics (Computer Science focus) – University of Hannover
MSc Applied Mathematics (Modeling) – Brunel University, London

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Certifications (Selection)
iSAQB-F Software Architect · Functional Safety ISO 26262 · Medical Norm ISO 13485 · Generative AI with GitHub Copilot

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Languages
German (native) · English (fluent) · Spanish (advanced)